

INTRINSIC RECONSTRUCTION OF WEBS FROM NONREDUCED MULTIPLICATION-FAILURE SCHEMES

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ABSTRACT. We prove that the abstract nonreduced multiplication-failure scheme of a cube-zero algebra with Hilbert function $(1, 4, 6)$ determines its relation web up to projective equivalence whenever the web is basepoint-free and its Jacobian quartic is smooth. No further genericity condition is required. The intrinsic deepest normal cone recovers both irreducible Pluecker components of the web, transported by one projective transformation. A contraction-kernel theorem computes the exterior support of the Jacobian component: a smooth quartic has support ten, whereas secant and tangent four-vectors have support at most eight. This excludes all exceptional recombinations; indeed three essential variables suffice. The support of the ambient rank-defect locus is contained in the inverse image of the binary-quartic subspace variety; no equality or ambient component classification is asserted.

The first nilpotent layer recovers the polarized Jacobian K3. The full nonreduced scheme separates the web classes not separated by that polarized surface. A separate technical supplement retains the full primary-boundary and relative-specialization theory. The reconstruction proof uses the nonreduced normal cone directly, without a higher-corank primary atlas or an exceptional-open hypothesis.

1. INTRODUCTION

A rank condition records where multiplication fails. The scheme structure of that rank condition records how it fails. In the cube-zero multiplication problem considered here this distinction is geometric: the reduction is a Schubert divisor, whereas successive nilpotent layers recover a polarized Jacobian K3 surface, contact divisors on a Segre quadric, vertex-primary corrections, and the higher-rank boundary on which those objects specialize. The full nonreduced scheme also recovers the original relation web throughout the smooth Jacobian locus.

The central point of this paper is that these layers are not artifacts of a chosen determinantal presentation. They are powers of the intrinsic nilradical of the failure scheme. The determinantal presentation is used to calculate them; the resulting graded schemes, their associated points, and their multiplication are intrinsic. The deepest normal cone supplies a second intrinsic construction: it retains two projective Pluecker components, whose recombination is unique for every web with smooth Jacobian. This gives the inverse problem a geometric endpoint rather than only a finite-ambiguity statement.

1.1. The multiplication failure scheme. Let V be a complex vector space of dimension $e \in \{3, 4\}$, put $p = e(e - 1)/2$, and let

$$R \subset \operatorname{Sym}^2 V$$

be an e -dimensional relation space. Set

$$S_R = \operatorname{Sym}^2 V/R, \quad B_R = \mathbb{C} \oplus V \oplus S_R,$$

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with multiplication induced by the quotient $\mathrm{Sym}^2 V \rightarrow S_R$ and with $(V \oplus S_R)S_R = 0$. On

$$X_R = \mathrm{Gr}(e, V \oplus S_R)$$

let $\mathcal{K}$ be the tautological subbundle. For every $m \geq 2$ the multiplication map defines the same zeroth-Fitting scheme

$$\hat{D}_R = V(\mathrm{Fitt}_0 \mathrm{coker}[\mathrm{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \rightarrow B_R \otimes \mathcal{O}]). \quad (1.1)$$

Its reduction is the Schubert divisor $\hat{\Delta}_R$ on which $\mathcal{K} \rightarrow V$ drops rank.

Regard R as a web of quadrics on $\mathbb{P}(V^*)$. On the invariant open G_e° used throughout, the web is basepoint-free and its Jacobian hypersurface

$$Y_R = V(\det(\partial q_i / \partial x_j)) \subset \mathbb{P}(V^*) \quad (1.2)$$

is smooth. For $e = 4$ it is a quartic K3 surface.

1.2. Recovering the original web. The central inverse theorem uses the full nonreduced scheme, including information at the deepest projection-corank stratum. The following conclusion holds without an additional genericity condition and is proved in Section 4.

Theorem 1.1 (Intrinsic web reconstruction). *For every pair $R, R' \in G_4^\circ$,*

$$\hat{D}_R \simeq \hat{D}_{R'} \implies R' = gR \text{ for some } g \in \mathrm{PGL}(V).$$

Consequently the abstract failure scheme determines B_R throughout the smooth Jacobian locus and separates every finite web-to-K3 packet contained in this locus.

The proof first recovers $\mathrm{Gr}(4, S_R)$ by iterated reduced singular loci. Its nonreduced normal cone identifies the two components of

$$\bigwedge^4 \mathrm{Sym}^2 V = \mathbb{S}_{(4,3,1,0)} V \oplus \mathbb{S}_{(5,1,1,1)} V.$$

The second component is the Jacobian covariant; the first is not retained by the quartic alone. The two projective components define a line in the Pluecker space. The contraction-kernel theorem (Lemma 4.10) computes the exact exterior support of the Jacobian component. For a smooth quartic that support has dimension ten. A secant or tangent four-vector on the Grassmannian has support at most eight. This excludes every second point and every infinitesimal ambiguity, so the line meets the Grassmannian in one reduced point. In fact three essential variables already suffice. The global tensor rulings of the normal cone exclude the transposition ambiguity of an isolated square-matrix cone. No extension of an abstract scheme isomorphism to the ambient Grassmannian is assumed.

The exceptional fibres of the two-component invariant are described in Theorem S4.1. A proper intersection of the component line with the Grassmannian has length at most two. Its second point can coincide with the first, lie at a discarded pure endpoint, or give a distinct candidate exchanged by a regular residual involution. A positive-dimensional fibre is exactly a Grassmann line specified by a three-plane inside a five-plane. The tangent condition $Qw_R \in \bigwedge^3 R \wedge W$ distinguishes the double-point case from a reduced secant pair. All these alternatives are excluded on G_4° by Theorem 4.13. The ambient obstruction is confined to the binary-Jacobian boundary, defined invariantly by the first catalecticant. Thus the smooth family has no residual secant involution, flag-line ambiguity, or ramification divisor to resolve. Outside that family the component-fibre classification is not promoted to a classification of abstract failure-scheme isomorphisms.

1.3. Intrinsic reconstruction. On the projection-corank-at-most-one locus let D_R be the restriction of $\widehat{D}_R$, let $\mathcal{N}_R$ be its nilradical, and let E_R be the annihilator scheme of the first nilpotent layer. The first-layer reconstruction is the following result proved in Section 5.

Theorem 1.2 (Polarized reconstruction). *For $R \in G_e^\circ$, put*

$$\mathcal{L}_R = \omega_{E_R} \otimes \mathcal{N}_R^{\otimes -(p+e-1)}.$$

Then

$$\bigoplus_{n \geq 0} H^0(E_R, \mathcal{L}_R^{\otimes n}) \simeq \bigoplus_{n \geq 0} H^0(Y_R, \mathcal{O}_{Y_R}(en))$$

as graded algebras. In particular, for $e = 4$ the abstract failure scheme determines the polarized quartic K3 $(Y_R, \mathcal{O}_{Y_R}(1))$.

1.4. Organization and relation to prior work. Sections 2 and 3 construct the algebra and its Fitting scheme. Section 4 contains the complete reconstruction argument: the deepest stratum, its oriented normal cone, the universal coefficient readout, one common projective transformation, and the contraction-kernel and exterior-support theorems. The scheme structure of the cone is used directly; no primary decomposition of its ideal is needed. Sections 5–7 compare this invariant with the polarized Jacobian surface and its moduli. Section 8 identifies the classical representation and secant tools and the precise additional contraction calculation.

The technical supplement is a separate reading object, not part of the proof of the main inverse theorem. It contains the full nilpotent-depth, primary-boundary, collision, relative-filtration, and exact-coordinate results, as well as the detailed Reye comparison and the ambient component-pencil classification. All mathematical statements and labels of those results are retained. Its rank-defect schemes are defined by actual minors; their support is only contained in the binary-Jacobian boundary. No complete embedded-primary atlas of the highest-corank residual layers is used or asserted.

Finite exact checks accompany the calculations, but do not formally verify the written structural proofs. The historically relevant Ballico paper [2] remains a specific documentary issue: its complete text has not been obtained, and no theorem-level nonanticipation conclusion is drawn from its title or later citations. This does not change the hypotheses or conclusion of the reconstruction theorem proved here.

2. QUADRATIC RELATIONS AND A RELATIVE RAMIFICATION FAMILY

Put $N = e(e+1)/2$ and $P = \mathbb{P}(V^*)$. Let $\mathcal{H} \subset V \otimes \mathcal{O}_P$ be the tautological hyperplane bundle. Contraction is normalized by $c_\alpha(vw) = \alpha(v)w + \alpha(w)v$.

Lemma 2.1 (Restriction and contraction). *There is a locally split exact sequence*

$$0 \longrightarrow \mathrm{Sym}^2 \mathcal{H} \longrightarrow \mathrm{Sym}^2 V \otimes \mathcal{O}_P \xrightarrow{c} V \otimes \mathcal{O}_P(1) \longrightarrow 0.$$

The cokernels of $\mathrm{Sym}^2 \mathcal{H} \rightarrow S_R \otimes \mathcal{O}_P$ and $R \otimes \mathcal{O}_P \rightarrow V \otimes \mathcal{O}_P(1)$ are naturally isomorphic. Their determinant ideals define the same degree- e hypersurface when nonzero.

Proof. Locally choose ℓ with $\alpha(\ell) = 1$. The decomposition $\mathrm{Sym}^2 V = \mathrm{Sym}^2 H \oplus H\ell \oplus \mathbb{C}\ell^2$ identifies contraction with zero, the identity, and multiplication by two on the three summands, respectively. The twist accounts for rescaling α . Both cokernels are the quotient by $R + \mathrm{Sym}^2 \mathcal{H}$; hence their zeroth Fitting ideals agree. Finally, $\det c_R$ is a section of $\det V \otimes (\det R)^* \otimes \mathcal{O}_P(e)$. $\square$

Proposition 2.2. *If R is basepoint-free, its quadratic morphism $\phi_R : P \rightarrow \mathbb{P}(R^*)$ is finite and surjective of degree 2^{e-1} . Its ramification Fitting scheme is Y_R in (1.2). Moreover,*

$$R \text{ is basepoint-free} \iff \gamma_R(HV) = S_R \text{ for every hyperplane } H \subset V. \quad (2.1)$$

Proof. The pullback of $\mathcal{O}(1)$ is $\mathcal{O}_P(2)$. A positive-dimensional projective fibre contains a curve on which this ample line bundle would be trivial. Thus the proper morphism is quasi-finite, and hence finite. Its image has dimension $e - 1$ and is the whole target. Intersecting $e - 1$ pulled-back hyperplanes computes its degree.

For $J = (\partial q_i / \partial x_j)$, Euler's identity reads $J(x)x = 2(q_1(x), \dots, q_e(x))^t$. On charts where a source coordinate and a target coordinate are nonzero, choose the radial vectors first. The radial differential is a unit; its quotient is the projective differential. Removing that invertible block identifies the two ramification Fitting ideals. The contraction matrix is J^t , up to bases, so Lemma 2.1 applies.

A functional on S_R vanishing on $\gamma_R(HV)$ pulls back to one vanishing on $R + HV$. If $H = \ker \alpha$, then $(HV)^\perp = \mathbb{C}\alpha^2$. Thus such a nonzero functional exists exactly when all $q \in R$ vanish at $[\alpha]$. $\square$

Proposition 2.3 (The universal smooth open). *The set G_e° is nonempty, open, and $\mathrm{PGL}(V)$ -invariant. The determinant construction over it is a relative effective Cartier divisor $\mathcal{Y} \subset G_e^\circ \times P$, proper and smooth over G_e° . It is obtained by descent from the frame bundle of the universal quotient, and its formation commutes with arbitrary base change. Each fibre is integral, with trivial canonical bundle; for $e = 4$ it also has $H^1(\mathcal{O}) = 0$.*

Proof. Fix a p -dimensional space S and set $T = \mathrm{Hom}(\mathrm{Sym}^2 V, S)$. Restriction over $T \times P$ to $\mathrm{Sym}^2 \mathcal{H}$ is a surjective evaluation map to the rank- p^2 bundle of square matrices. The incidence of corank at least two has codimension four on each fibre over P , hence dimension $\dim T + e - 5 < \dim T$. Its projection to T is proper and has proper closed image. Similarly, HV has dimension $p + e - 1$; nonsurjectivity of its restriction has codimension e and universal dimension $\dim T - 1$. Remove these two images and nonsurjective tensors. Also require invertibility of the square restriction at one fixed hyperplane. Each is a nonempty open condition on the irreducible space T , so their intersection T_1 is nonempty.

On $T_1 \times P$ the universal determinant section is nonzero on every fibre. Since a nonzero homogeneous polynomial is a non-zero-divisor on projective space over a field, its zero scheme Z is a relative Cartier divisor and is flat over T_1 . More explicitly, apply the fibrewise flatness criterion to $0 \rightarrow \mathcal{O}(-e) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_Z \rightarrow 0$: the first two terms are base-flat and the first map stays injective on every fibre. This also proves that the same sequence stays exact after any base change.

The total space Z is smooth over $\mathbb{C}$. Indeed, evaluation is a submersion in the tensor directions, and the determinant hypersurface of square matrices is smooth at matrices of corank one; the corank-two incidence was removed. Every fibre is a nonempty degree- e hypersurface. The generic fibre over T_1 is regular, since its local rings are localizations of the regular local rings of Z . In characteristic zero it is geometrically regular, hence smooth. The nonsmooth locus of $Z \rightarrow T_1$ is closed, and its image is closed by properness and misses the generic point. Deleting that image gives a nonempty open with smooth fibres.

Surjections $\mathrm{Sym}^2 V \twoheadrightarrow S$ form the frame bundle of the rank- p universal quotient on $\mathrm{Gr}(e, \mathrm{Sym}^2 V)$. The conditions of basepoint-freeness and smooth determinant fibre depend only on the kernel. Their bad loci have closed proper projections from the corresponding projective incidences. They therefore define the asserted open downstairs. The determinant ideal, relative Cartier property, and flatness descend along the faithfully flat frame bundle; flatness with smooth geometric fibres gives smoothness. The cokernel and Fitting constructions themselves commute with arbitrary base change. The extra fixed-hyperplane condition above is used only to produce a nonempty open in T ; it is not imposed in the definition downstairs.

A smooth positive-degree hypersurface in $\mathbb{P}^{e-1}$ is integral here: distinct positive-degree components would meet and give a singular point, and repeated components are not smooth. Adjunction gives $\omega_{Y_R} = \mathcal{O}_{Y_R}$. The hypersurface sequence gives genus one when $e = 3$ and $H^1(Y_R, \mathcal{O}) = 0$

when $e = 4$. Thus the latter is a K3 surface. Smoothness of the determinant also excludes corank at least two in the square restriction, since all cofactors then vanish. $\square$

Lemma 2.4. *The algebras B_R and $B_{R'}$ are isomorphic exactly when R and R' lie in the same $\mathrm{PGL}(V)$ -orbit.*

Proof. An isomorphism preserves the maximal ideal and its square. On their associated graded spaces it induces an isomorphism of V carrying the kernel of quadratic multiplication to the other kernel. Conversely such a linear isomorphism gives an isomorphism of the graded algebras. Scalar matrices act trivially on relation subspaces. $\square$

3. THE MULTIPLICATION IDEAL AND INTRINSIC TORSION

We first calculate on the whole Grassmannian. In this section $R \in G_e^\circ$ is fixed, and the subscript R is omitted.

Proposition 3.1 (The full Fitting presentation). *On a frame chart write $\mathcal{K} \rightarrow V \oplus S$ as $\begin{pmatrix} M \\ A_0 \end{pmatrix}$. For every $m \geq 2$,*

$$\mathcal{I}_{\widehat{D}} = (\det M)I_p(\gamma \mathrm{Sym}^2 M). \quad (3.1)$$

The reduction of $\widehat{D}$ is the integral divisor $\widehat{\Delta}$. Its smooth locus is the rank- $e - 1$ stratum. Consequently D is the restriction of $\widehat{D}$ to the smooth locus of its reduction, and this restriction is intrinsic.

Proof. Cube-zero multiplication and the unit show, over the coefficient ring and not only on fibres, that the image of every degree $m \geq 2$ is $\mathbb{C}1 + \mathcal{K} + \mathcal{K}^2$. Its presentation is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \mathrm{Sym}^2 M \end{pmatrix}.$$

Every nonzero maximal minor uses the scalar column and all e columns meeting the V -rows. Expanding on these rows proves (3.1). The graph block A_0 has no effect on this ideal. Off $\det M = 0$, the quadratic block is surjective because γ is surjective, so the ideal is the unit ideal. On the divisor it vanishes. The divisor is the usual irreducible Schubert divisor: the rank- $e - 1$ stratum is dense and integral, and every lower-rank plane is a specialization of that stratum. Equivalently the standard Schubert charts give its reduced determinantal equations.

Near a point with projection rank r , an invertible r -minor reduces the projection to $\mathrm{diag}(I_r, T)$, where the entries of the $(e - r) \times (e - r)$ block T are independent transverse coordinates. One can see their independence by varying the V -components of a basis of $K \cap S$ while keeping an invertible S -frame block fixed. The determinant equation is $\det T$. Its gradient is nonzero exactly at residual corank one. This proves the smooth-locus claim, including its assertion on the Grassmannian rather than on a chosen frame space. $\square$

On Δ put $H = \mathrm{im}(K \rightarrow V)$ and $\Lambda = K \cap S$. These give a morphism to $P \times \mathbb{P}(S)$. Define $E = \Delta \times_P Y$. The graph of K/Λ yields isomorphisms

$$\Delta \simeq \mathrm{Tot}_{P \times \mathbb{P}(S)} \mathrm{Hom}(\mathcal{H}, S/\Lambda), \quad (3.2)$$

$$E \simeq \mathrm{Tot}_{Y \times \mathbb{P}(S)} \mathrm{Hom}(\mathcal{H}, S/\Lambda). \quad (3.3)$$

Indeed the inverse construction takes the inverse image of a graph $H \rightarrow S/\Lambda$ in $H \oplus S$. Both constructions commute with base change. In particular E and Δ are smooth integral, of dimensions $ep - 2$ and $ep - 1$.

Theorem 3.2 (Ramification normal form). *On all of U the ideal of D is*

$$\mathcal{I}_D = \mathcal{I}_\Delta \mathcal{I}_E = \mathcal{I}_\Delta \cap \mathcal{I}_E^2. \quad (3.4)$$

The associated supports are exactly Δ and E . The nilradical is the invertible $\mathcal{O}_E$ -module $\mathcal{N} = \mathcal{O}_E(-\Delta)$ and is square-zero. Its annihilator recovers E as a scheme.

Proof. Near projection corank one, Gaussian operations make $M = \text{diag}(I_{e-1}, t)$. The quadratic block becomes $[A, tB, t^2c]$, with $A|_{t=0} = \gamma|_{\text{Sym}^2 H}$. Equation (2.1) says $[A, B]$ is onto locally; choose $AU + BV = I_p$. Multiplication by t then shows that $[A, tB, t^2c]$ and $[A, tI_p]$ have exactly the same column module. Their zeroth Fitting ideals agree. Off Y , A is invertible and (3.1) is (t) . On Y , the restriction has rank $p-1$ by Proposition 2.3. A Schur complement gives a residual row with ideal (t, f) , where $f \bmod t$ is a local ramification equation. Thus the full ideal is

$$(t^2, tf) = t(t, f). \quad (3.5)$$

The pair t, f is a regular parameter pair cutting out E . The identity $(t) \cap (t, f)^2 = t(t, f)$ proves (3.4). These are primary ideals because their associated graded rings are polynomial rings over the integral smooth supports. At the generic point of E , neither primary component can be omitted. Their radicals are distinct, giving exactly the stated associated points. Finally the nilradical is $(t)/t(t, f)$; cancellation of t gives its module identification and annihilator $(t, f)/t(t, f)$. Both are intrinsic, so the calculation glues. $\square$

Lemma 3.3 (Generic torsion as a sheaf). *Let Z be noetherian with integral reduction and generic point η . For a coherent module $\mathcal{F}$ define*

$$\text{tor}_\eta(\mathcal{F}) = \ker(\mathcal{F} \longrightarrow i_* \widetilde{\mathcal{F}}_\eta), \quad i : \text{Spec } \mathcal{O}_{Z, \eta} \longrightarrow Z.$$

This kernel is coherent and is the largest coherent submodule with zero generic stalk. It is preserved by isomorphisms of schemes.

Proof. On an affine open meeting η this is the localization kernel at the prime of η . It is a submodule of a finite module over a noetherian ring, hence finite. Localization of that kernel on smaller affine opens is the corresponding kernel, so these modules glue. Every submodule with zero generic stalk lies in the kernel, and the kernel itself has zero generic stalk. The definition involves no ambient embedding or chosen primary decomposition. $\square$

4. THE DEEPEST NORMAL CONE AND RECONSTRUCTION OF THE WEB

The polarized Jacobian retains only one of two irreducible components of the Pluecker vector of a web. The full failure scheme retains both. In this section we recover these two projective components intrinsically and prove that they determine every web in the smooth Jacobian open. Exterior support excludes every exceptional pencil in this geometric family. Thus the deeper boundary separates the finite ambiguity of the polarized-K3 invariant. The normal cone used here is along the deepest Schubert stratum, not along the reduction of the failure scheme.

Throughout this section $\dim V = 4$, $W = \text{Sym}^2 V$, and $E = \bigwedge^4 W$. There is a multiplicity-free decomposition

$$E = E_{175} \oplus E_{35}, \quad E_{175} = \mathbb{S}_{(4,3,1,0)} V, \quad E_{35} = \mathbb{S}_{(5,1,1,1)} V \simeq \det V \otimes \text{Sym}^4 V. \quad (4.1)$$

Write P and Q for its two equivariant projections, in that order. For a web R , write $[w_R] \in \mathbb{P}(E)$ for its Pluecker point. On the locus where both projections are nonzero, put

$$\mathcal{L}_R = \mathbb{P}(\mathbb{C}Pw_R \oplus \mathbb{C}Qw_R).$$

The quadratic ideal of $\text{Gr}(4, W)$ is denoted by $\mathcal{I}_{\text{Pl}, 2}$. Each of its quadrics satisfies

$$F(aPw_R + bQw_R) = (b - a)\ell_{F, R}(a, b), \quad F \in \mathcal{I}_{\text{Pl}, 2}, \quad (4.2)$$

for a linear form $\ell_{F,R}$, since w_R is decomposable. Put $X = \text{Gr}(4, W)$, and let $X^\times$ be the open where both projections are nonzero. Set $H_w = \langle \ell_{F,w} : F \in \mathcal{I}_{\text{Pl},2} \rangle$, and let $\pi : X^\times \rightarrow \mathbb{P}(E_{175}) \times \mathbb{P}(E_{35})$ send $[w]$ to its pair of component lines. Define $G_4^{\text{rec}} \subset G_4^\circ$ by the conditions

$$Pw_R \neq 0, \quad Qw_R \neq 0, \quad \dim \langle \ell_{F,R} : F \in \mathcal{I}_{\text{Pl},2} \rangle = 2. \quad (4.3)$$

These conditions are independent of a representative of $[w_R]$ and invariant under $\text{PGL}(V)$.

Theorem 4.1 (Intrinsic reconstruction on the smooth Jacobian locus). *The reconstruction open is the whole smooth geometric locus: $G_4^{\text{rec}} = G_4^\circ$. Let $R \in G_4^\circ$ and $R' \in G_4^\circ$. Every isomorphism of complex schemes*

$$\widehat{D}_R \simeq \widehat{D}_{R'}$$

implies $R' = gR$ for some $g \in \text{PGL}(V)$. In particular, the abstract nonreduced failure scheme determines the cube-zero algebra B_R , up to isomorphism, throughout G_4° .

More precisely, the normal cone of $\widehat{D}_R$ along its intrinsically determined deepest Schubert stratum recovers $([Pw_R], [Qw_R])$, up to the same projective change of coordinates in V . The webs with this pair of projective components are exactly

$$(\mathcal{L}_R \cap \text{Gr}(4, W)) \setminus \{[Pw_R], [Qw_R]\}. \quad (4.4)$$

On G_4^{rec} , this intersection is the single reduced point $[w_R]$, before taking projective-coordinate orbits.

4.1. The intrinsic deepest stratum and its oriented tensor cone. For clarity, set

$$\Sigma_r(R) = \{K \in \text{Gr}(4, V \oplus S_R) : \text{rank}(K \rightarrow V) \leq r\}, \quad 0 \leq r \leq 3,$$

with reduced structure. Thus $\Sigma_3(R) = \widehat{\Delta}_R$ and $\Sigma_0(R) = \text{Gr}(4, S_R)$.

Lemma 4.2 (The deepest normal cone). *Starting with the reduction of $\widehat{D}_R$, three iterations of taking the reduced singular locus give $\Sigma_0(R)$. Consequently $\Sigma_0(R)$ and $\mathcal{C}_R = C_{\Sigma_0(R)/\widehat{D}_R}$ are intrinsic. Let $\mathcal{K}_0$ denote the tautological rank-four bundle on $\Sigma_0(R)$. The ambient vector bundle determined by the degree-one part is $\text{Hom}(\mathcal{K}_0, V)$. At a point $K_0 \subset S_R$, choose bases and write $T : K_0 \rightarrow V$. The fibre cone is*

$$\text{Spec } \mathbb{C}[t_{ij}] / (d(T)J_R(T)), \quad d(T) = \det T, \quad J_R(T) = I_6(\gamma_R \text{Sym}^2 T). \quad (4.5)$$

Its ideal is generated in degree sixteen and its reduction is the determinant hypersurface. The two rulings of its rank-one locus are intrinsically distinguishable over $\Sigma_0(R)$: the V -ruling is the projectively trivial one.

Proof. At a point of projection rank r , invert an r -minor and take the Schur complement. The remaining $(4-r)^2$ entries are independent coordinates and the rank conditions become the usual matrix rank conditions, times smooth auxiliary coordinates. The Jacobian criterion for minors gives

$$(\text{Sing } \Sigma_r(R))_{\text{red}} = \Sigma_{r-1}(R), \quad 1 \leq r \leq 3.$$

This also follows directly by differentiating the minors: the rank- r stratum is smooth, whereas at rank at most $r-1$ the tangent dimension exceeds the local dimension. Since reduction and reduced singular locus are preserved by abstract isomorphisms, $\Sigma_0(R)$ is intrinsic.

Choose a splitting $S_R = K_0 \oplus S'$. A Grassmann chart near K_0 consists of graphs

$$(T, B) : K_0 \longrightarrow V \oplus S'.$$

The deepest stratum is $T = 0$; B provides its eight smooth coordinates. The Fitting factorization gives precisely $d(T)J_R(T)$, independently of B . All generators of J_R have degree twelve in T , so passing to the (t_{ij}) -adic associated graded gives the same homogeneous ideal. This proves (4.5). There are no degree-one relations. The dual of the conormal bundle is therefore $\text{Hom}(\mathcal{K}_0, V)$, as

also follows by quotienting the Grassmannian tangent bundle along $\text{Gr}(4, S_R)$. If T is invertible, $\gamma_R \text{Sym}^2 T$ is onto, so its maximal minors do not vanish simultaneously. Thus $V(dJ_R) = V(d)$; the determinant is irreducible, and the Nullstellensatz gives $\sqrt{dJ_R} = (d)$. This argument does not require any higher-colon or primary-boundary calculation.

The rank-one subcone is recovered from the reduced determinant cone by iterated reduced singular loci in each fibre. Its projectivization is the relative Segre variety

$$\mathbb{P}(V) \times \mathbb{P}(\mathcal{K}_0^*) \longrightarrow \Sigma_0(R).$$

The two components of the relative parameter space of its maximal linear $\mathbb{P}^3$'s are $\mathbb{P}(V) \times \Sigma_0(R)$ and $\mathbb{P}(\mathcal{K}_0^*)$. To see that these are exactly the two components, note that a vector space of rank-one tensors cannot contain both $v_1 \otimes u_1$ and $v_2 \otimes u_2$ with both pairs independent: their sum has rank two. Thus a maximal such space fixes one tensor factor.

The first parameter bundle is projectively trivial. The second is not. Restrict to a Schubert line in $\text{Gr}(4, 6)$, obtained by varying K_0 between a fixed three-plane and five-plane. On this line

$$\mathcal{K}_0^*|_{\mathbb{P}^1} \simeq \mathcal{O}_{\mathbb{P}^1}^{\oplus 3} \oplus \mathcal{O}_{\mathbb{P}^1}(1).$$

If its projectivization were a trivial $\mathbb{P}^3$ -bundle, the vector bundle would be a direct sum of four copies of one line bundle. This follows by comparing the relative $\mathcal{O}(1)$ on a projective-bundle isomorphism and pushing forward. Its determinant degree would then be divisible by four, contrary to the degree one displayed above. Triviality is invariant under an isomorphism of the base. Since the base is connected, an isomorphism of the cone cannot interchange these two parameter bundles on some fibres but not others. It must preserve the V -ruling. $\square$

Remark 4.3 (Why the ruling argument is needed). A single determinant cone of square matrices permits transposition. The preceding relative argument removes that ambiguity using only intrinsic data on $\Sigma_0(R)$; it does not assume that an isomorphism of failure schemes extends to the ambient Grassmannian. A linear isomorphism of tensor spaces preserving the two rank-one rulings has the form $g \otimes h$. Indeed, its restrictions to the two Segre factors are projective linear, and after lifting those maps the remaining linear map fixes every decomposable line. Applying it to sums with one common factor shows that its scalar is constant. Thus the coordinate change in the recovered V factor is an ordinary projective linear change, as required below.

4.2. The universal Schur coefficient map. The distinction between an isotypic component and a specified line in its multiplicity space is essential here. We establish the coefficient identity before fixing the web. All vector spaces in this subsection are complex and four-dimensional. Put

$$\mathcal{F}(X) = \bigwedge^6 \text{Sym}^2 X, \quad \Lambda = \{(5, 4, 2, 1), (4, 4, 4, 0)\}.$$

For a linear map T write $\mathcal{F}(T) = \bigwedge^6 \text{Sym}^2 T$.

Lemma 4.4 (Universal coefficient identity). *The natural degree-twelve map*

$$\begin{aligned} \Phi_{V,U} : \mathcal{F}(V)^* \otimes \mathcal{F}(U) &\longrightarrow \text{Sym}^{12}(V^* \otimes U), \\ \Phi_{V,U}(\beta \otimes \xi)(T) &= \beta(\mathcal{F}(T)\xi) \end{aligned} \tag{4.6}$$

is $GL(V) \times GL(U)$ -equivariant. Under the functorial multiplicity-free decompositions

$$\mathcal{F}(X) = \mathbb{S}_{(5,4,2,1)} X \oplus \mathbb{S}_{(4,4,4,0)} X \tag{4.7}$$

and the Cauchy decomposition

$$\text{Sym}^{12}(V^* \otimes U) = \bigoplus_{\substack{\lambda \vdash 12 \\ \ell(\lambda) \leq 4}} \mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U,$$

the restriction of $\Phi_{V,U}$ to $\mathbb{S}_\lambda V^* \otimes \mathbb{S}_\mu U$ is zero if $\lambda \neq \mu$. For $\lambda = \mu \in \Lambda$, it is a nonzero scalar multiple of the canonical Cauchy inclusion. With compatible functorial normalizations this scalar is one. Consequently, for every fixed $\beta = \sum_\lambda \beta_\lambda$,

$$\Phi_{V,U}(\mathbb{C}\beta \otimes \mathcal{F}(U)) = \bigoplus_{\lambda \in \Lambda} \mathbb{C}\beta_\lambda \otimes \mathbb{S}_\lambda U, \quad (4.8)$$

where each tensor product on the right is placed in its Cauchy summand. A zero component contributes the zero subspace.

Proof. The action on matrices is $T \mapsto gTh^{-1}$; the action on functions is $f(T) \mapsto f(g^{-1}Th)$. Functoriality of $\mathcal{F}$ shows directly that (4.6) intertwines this action with $\beta \otimes \xi \mapsto (\mathcal{F}(g^{-1})^*\beta) \otimes \mathcal{F}(h)\xi$. Thus equivariance is a statement about the universal map, not about a fixed- β slice.

For completeness, the exterior character identity used here is

$$[z^6] \prod_{1 \leq i \leq j \leq 4} (1 + zx_i x_j) = s_{(5,4,2,1)}(x) + s_{(4,4,4,0)}(x).$$

Equivalently, it follows by exterior duality from (4.1) and $\det(\text{Sym}^2 X) = (\det X)^5$. The two dimensions are 175 and 35, whose sum is $\binom{10}{6}$. The characteristic-zero Schur decomposition is functorial; the Cauchy formula is multiplicity-free as a representation of the product group [18, 8]. The source of Φ is the direct sum of the four irreducible product representations indexed by $\Lambda \times \Lambda$. Only the two diagonal ones occur in the target. Schur's lemma kills the off-diagonal blocks and makes each diagonal block a scalar multiple of its Cauchy inclusion. This also proves that no other left multiplicity line can enter the formula.

Here is an explicit nonvanishing check for both scalars. For an ordered basis $x_1, \dots, x_4$, order the monomials of $\text{Sym}^2 X$ as

$$(b_1, \dots, b_{10}) = (x_1^2, x_1 x_2, x_1 x_3, x_1 x_4, x_2^2, x_2 x_3, x_2 x_4, x_3^2, x_3 x_4, x_4^2).$$

Highest-weight vectors for the two summands are

$$\xi_{5421} = b_1 \wedge b_2 \wedge b_3 \wedge b_4 \wedge b_5 \wedge b_6, \quad \xi_{4440} = b_1 \wedge b_2 \wedge b_3 \wedge b_5 \wedge b_6 \wedge b_8.$$

Their weights are respectively $(5, 4, 2, 1)$ and $(4, 4, 4, 0)$. Every raising operator $x_i \partial / \partial x_j$, $i < j$, either gives zero or repeats a wedge factor, so both are highest vectors. Complete reducibility and the dimension sum above show that these two generated irreducible submodules exhaust $\mathcal{F}(X)$; this also proves (4.7) directly. Choose β_λ in the dual summand with $\beta_\lambda(\xi_\lambda) = 1$, and extend it by zero on the other summand. After identifying the ordered bases of U and V , evaluation at $T = \text{diag}(t_1, t_2, t_3, t_4)$ gives

$$\begin{aligned} \Phi(\beta_{5421} \otimes \xi_{5421})(T) &= t_1^5 t_2^4 t_3^2 t_4, \\ \Phi(\beta_{4440} \otimes \xi_{4440})(T) &= t_1^4 t_2^4 t_3^4. \end{aligned} \quad (4.9)$$

In particular both evaluations are one at the identity. This proves nonvanishing without a computed rank or a choice of numerical web. The canonical coefficient realization of the Cauchy inclusion is $\beta \otimes \xi \mapsto [T \mapsto \beta(\mathbb{S}_\lambda(T)\xi)]$. Compatible functorial splittings identify the restriction of $\mathcal{F}(T)$ with $\mathbb{S}_\lambda(T)$, giving scalar one; other scalar normalizations have the same image. Finally let ξ vary independently in its two summands. The off-diagonal vanishing and the nonzero diagonal scalars give (4.8). $\square$

Lemma 4.5 (Exterior duality and the common determinant twist). *Set $W = \text{Sym}^2 V$. The canonical isomorphism*

$$\begin{aligned} \bigwedge^4 W \otimes (\det W)^* &\longrightarrow \bigwedge^6 W^*, \\ w \otimes \omega &\longmapsto [\eta \mapsto \omega(w \wedge \eta)] \end{aligned} \quad (4.10)$$

sends the Pluecker line of $R = \ker \gamma_R$ to the line of $\beta_R = \gamma_R^*(\text{vol}_{S_R})$. It identifies $E_{175} \otimes (\det V)^{-5}$ with $\mathbb{S}_{(5,4,2,1)} V^*$, and $E_{35} \otimes (\det V)^{-5}$ with $\mathbb{S}_{(4,4,4,0)} V^*$.

Proof. Extend a basis of R to a basis of W . Both sides of the first assertion are nonzero alternating six-forms annihilating a wedge containing a vector of R , and on the six complementary basis vectors they agree after one scalar normalization. This proves the line identity. The map in (4.10) is natural for all isomorphisms of W . The highest weights of the two twisted summands are $(-1, -2, -4, -5)$ and $(0, -4, -4, -4)$, respectively: these are the negatives of the reversed partitions $(5, 4, 2, 1)$ and $(4, 4, 4, 0)$. Thus both labels and the common twist are as stated. Changing the volume multiplies both components by the same scalar. $\square$

4.3. Reading both Pluecker components from the residual ideal.

Lemma 4.6 (The two-component readout). *At any fibre of the oriented cone in Lemma 4.2, its degree-sixteen ideal recovers the projective pair $([Pw_R], [Qw_R])$ whenever both components are nonzero. It detects the vanishing of either component as well.*

Proof. The reduced cone determines its determinant equation d up to a nonzero scalar. The graded algebra determines the homogeneous ideal in the symmetric algebra of degree one. Multiplication by d in that polynomial ring is injective; dividing the degree-sixteen ideal by d therefore recovers precisely $(J_R)_{12}$. This uses the nonreduced ideal, not just its support.

Let $U = K_0$ and choose a volume on S_R . Its pullback $\beta_R \in \Lambda^6(\text{Sym}^2 V)^*$ pairs with $\mathcal{F}(T)$ to give exactly the maximal minors of $\gamma_R \text{Sym}^2 T$. Lemma 4.4 now gives, as an equality of subspaces in the Cauchy decomposition,

$$(J_R)_{12} = \bigoplus_{\lambda \in \Lambda} \mathbb{C} \beta_{R,\lambda} \otimes \mathbb{S}_\lambda U. \quad (4.11)$$

This is the fixed-slice consequence of the universal block identity. It does not treat the fixed slice as a $GL(V)$ -representation. Intersecting with a Cauchy summand and contracting its right factor with all elements of $(\mathbb{S}_\lambda U)^*$ recovers exactly $\mathbb{C} \beta_{R,\lambda}$, including its possible vanishing. Lemma 4.5 identifies these two lines with $[Pw_R]$ and $[Qw_R]$, with their labels and one common determinant twist. Right coordinate changes preserve the left lines; left coordinate changes act on both by the same representation of the coordinate transformation. Naturality under an abstract isomorphism is isolated in Lemma 4.7 below. $\square$

4.4. Naturality under an abstract isomorphism.

Lemma 4.7 (One projective transformation for both coefficient lines). *Let $\varphi : \widehat{D}_R \xrightarrow{\sim} \widehat{D}_{R'}$ be an abstract isomorphism, with $R, R' \in G_4$. It induces an isomorphism $\sigma : \Sigma_0(R) \rightarrow \Sigma_0(R')$ and a grading-preserving isomorphism of their normal cones. There is one $g \in \text{PGL}(V)$, independent of the point of $\Sigma_0(R)$, which transports both recovered nonzero Pluecker component lines:*

$$[Pw_{R'}] = [P((\bigwedge^4 \text{Sym}^2 g)w_R)], \quad [Qw_{R'}] = [Q((\bigwedge^4 \text{Sym}^2 g)w_R)].$$

The corresponding vanishing statements hold for zero components. No extension of φ to either ambient Grassmannian is required.

Proof. The deepest stratum is intrinsic by Lemma 4.2. If $\mathcal{I}$ and $\mathcal{I}'$ are its ideals in the two failure schemes, pullback by φ identifies every power of these ideals. Hence it induces the graded algebra isomorphism

$$\sigma^* \left(\bigoplus_{m \geq 0} \mathcal{I}^m / \mathcal{I}^{m+1} \right) \simeq \bigoplus_{m \geq 0} \mathcal{I}'^m / \mathcal{I}'^{m+1}. \quad (4.12)$$

The symmetric algebra of degree one surjects canonically onto either algebra. Thus (4.12) also identifies its homogeneous kernel, not merely the cone's reduction; this is the functorial normal-cone construction [19]. The degree-one part recovers the vector bundle $\text{Hom}(\mathcal{K}_0, V)$.

The relative rank-one Segre variety and its two ruling parameter bundles are preserved. Their distinction by projective triviality was proved in Lemma 4.2; in particular the V -ruling cannot be exchanged with the other ruling. The induced isomorphism of the trivial $\mathbb{P}(V)$ -bundles, after pullback by σ , defines a morphism $\Sigma_0(R) \rightarrow \text{PGL}(V)$. The Grassmannian $\Sigma_0(R)$ is connected and projective and has only constant global regular functions, whereas $\text{PGL}(V)$ is affine. This morphism is therefore constant. Denote its value by g .

Fix a point $U = K_0$ and its image $U' = K'_0$. A linear isomorphism preserving both rank-one rulings has, after a lift of g , the form $\alpha(T) = gTh^{-1}$ for some isomorphism $h : U \rightarrow U'$. Indeed the two Segre factors give projective linear maps. After lifting them, the remaining linear map fixes each decomposable line; sums of tensors with a common factor force its scalar to be constant. This scalar can be absorbed in h .

The coefficient map has the following commutative square:

$$\begin{array}{ccc} \mathcal{F}(V)^* \otimes \mathcal{F}(U') & \xrightarrow{\Phi_{V,U'}} & \text{Sym}^{12}(V^* \otimes U') \\ \mathcal{F}(g)^* \otimes \mathcal{F}(h^{-1}) \downarrow & & \downarrow \alpha^* \\ \mathcal{F}(V)^* \otimes \mathcal{F}(U) & \xrightarrow{\Phi_{V,U}} & \text{Sym}^{12}(V^* \otimes U). \end{array}$$

In fact both routes evaluate at T as $\beta'(\mathcal{F}(g)\mathcal{F}(T)\mathcal{F}(h^{-1})\xi')$. The scalar determinant equation of the reduced cone is transported by α . Division of the transported degree-sixteen ideal by that equation consequently transports $(J_{R'})_{12}$ to $(J_R)_{12}$. The universal diagonal-block identity gives

$$\mathbb{C}\beta_{R,\lambda} = \mathbb{C}\mathcal{F}(g)^*\beta_{R',\lambda} \quad (\lambda \in \Lambda)$$

in the respective left factors; h acts only on the right factor and cannot change either left line. Exterior duality in Lemma 4.5 now gives the displayed Pluecker identities. Its determinant multiplier is common to both summands and disappears projectively. Every step arose from the isomorphism of pairs, establishing the last assertion. $\square$

4.5. Exterior support of the Jacobian component. The obstruction to recombining two projective Schur components has a simple linear-algebraic description. A secant or tangent vector of $\text{Gr}(4, W)$ uses at most eight directions in W . The Jacobian component of a smooth quartic uses all ten. We prove a sharper statement which also applies to singular quartics.

Fix a volume $\epsilon = x_1 \wedge x_2 \wedge x_3 \wedge x_4$ and suppress it when writing quartics. Define the equivariant map

$$j : \det V \otimes \text{Sym}^4 V \longrightarrow \bigwedge^4 \text{Sym}^2 V$$

by polarization of

$$j(\epsilon \otimes \ell^4) = (\ell x_1) \wedge (\ell x_2) \wedge (\ell x_3) \wedge (\ell x_4). \quad (4.13)$$

The right side changes by the determinant of a change of the x_i , so this definition does not depend on the chosen volume. More explicitly,

$$j(\epsilon \otimes v_1 v_2 v_3 v_4) = \frac{1}{24} \sum_{\sigma \in \mathfrak{S}_4} (v_{\sigma(1)} x_1) \wedge \cdots \wedge (v_{\sigma(4)} x_4). \quad (4.14)$$

This is nonzero. Its irreducible source and (4.1) identify its image with E_{35} . For the Jacobian map C , evaluation at $\ell = x_1$ gives $Cj(x_1^4) = 2x_1^4$; hence

$$Cj = 2 \text{id}, \quad Q = \frac{1}{2}jC. \quad (4.15)$$

These identities also give $j = C^*/24$ in the Fischer normalization used below.

For $q \in W^* = \text{Sym}^2 V^*$, regard q as the symmetric bilinear functional $q(uv)$, and let ι_q denote exterior contraction. A quartic has *essential-variable space* $H \subset V$ if H is the smallest subspace for which it lies in $\text{Sym}^4 H$. Equivalently, $H^\perp$ is the kernel of its first differentiation map $V^* \rightarrow \text{Sym}^3 V$. This is the first-catalecticant characterization of essential variables [31, Proposition 1]; it also follows by choosing coordinates along the kernel and using characteristic zero. This subspace of V is distinct from the exterior-support subspace of $W = \text{Sym}^2 V$ defined below.

4.5.1. Two representation-theoretic lemmas. We keep the determinant line $D = \det V$ in the notation. The first lemma concerns a polynomial representation with its actual weight decomposition, not just its associated graded vector space.

Lemma 4.8 (Shear descent in a fixed weight). *Let $V = A \oplus B$, with $B \neq 0$, and let $m \geq 1$. Suppose that $N \subset D \otimes \text{Sym}^m V$ is invariant under $\text{id}_A \oplus t \text{id}_B$, $t \in \mathbb{C}^*$, and under all shears*

$$u_L(a + b) = a + b + L(a), \quad L \in \text{Hom}(A, B).$$

If $N \not\subset D \otimes \text{Sym}^m B$, then

$$N \cap (D \otimes A \otimes \text{Sym}^{m-1} B) \neq 0.$$

More precisely, a nonzero vector in the A -degree- k weight space, $k > 0$, has a nonzero image in the degree-one space under a product of $k - 1$ infinitesimal rank-one shears. The same fixed nonzero vector of B can be used in every shear.

Proof. Put $s = \dim B$. The displayed torus acts on $D \otimes \text{Sym}^k A \otimes \text{Sym}^{m-k} B$ with weight $s + m - k$. These weights are distinct. Since a complex torus acts semisimply, its invariant subspace N is the direct sum of its intersections with these weight spaces. This justifies projecting a vector of N to one degree before differentiating; no cancellation between degrees is possible.

Choose a basis $a_1, \dots, a_r$ of A , a nonzero $b \in B$, and a basis vector ϵ of D . Write a nonzero degree- k vector as $\epsilon \otimes F$, where

$$F = \sum_{|\beta|=k} a^\beta f_\beta, \quad f_\beta \in \text{Sym}^{m-k} B.$$

Choose β with $f_\beta \neq 0$, and i with $\beta_i > 0$. Set $\alpha = \beta - e_i$. In $\partial_A^\alpha F$, the coefficient of a_i is exactly $\beta! f_\beta$. Indeed a monomial that contributes to this coefficient must have exponent $\alpha + e_i = \beta$. This coefficient is nonzero in characteristic zero, so the derivative is a nonzero element of $A \otimes \text{Sym}^{m-k} B$.

The shear sending a_j to $a_j + tb$ and fixing the other basis vectors has determinant one. Its infinitesimal action on $D \otimes \text{Sym}^m V$ is consequently $1_D \otimes b \partial_{a_j}$, with no extra action on D . Invariance under the shear implies invariance under its infinitesimal action: the orbit is a polynomial in t , and all its coefficients belong to the invariant linear subspace. These differential operators commute, because every ∂_{a_j} kills b . Their product with multiplicities α_j sends our vector to

$$\epsilon \otimes b^{k-1} \partial_A^\alpha F.$$

This vector belongs to N . It is nonzero, since multiplication by b^{k-1} is injective in the polynomial algebra $\text{Sym} B$. Its exact bidegree is $(1, m - 1)$; thus its image in the corresponding weight-graded piece is also nonzero. The proof uses neither a special choice of b nor a restriction on r or s . When $k = 1$ the product of shears is the identity. This proves all assertions, including the low-dimensional cases. $\square$

For the next lemma write $[a, b] = \text{Sym}^a U \otimes \text{Sym}^b U'$, where U, U' are the two standard two-dimensional representations of $SL_2 \times SL_2$. Only even weights occur below, so all representations descend through its double cover of SO_4 .

Lemma 4.9 (The exterior-cube character). *Let q be nondegenerate on the four-dimensional space V , and put $W = \text{Sym}^2 V$. Under $SL_2 \times SL_2 \rightarrow SO(V, q)$, choose $V = [1, 1]$. Then*

$$\begin{aligned} \bigwedge^3 W \simeq & [6, 0] \oplus [0, 6] \oplus [4, 4] \oplus 2[4, 2] \oplus 2[2, 4] \\ & \oplus [2, 2] \oplus 2[2, 0] \oplus 2[0, 2]. \end{aligned} \quad (4.16)$$

In particular $(\bigwedge^3 W)^{SO(V, q)} = 0$, and $\text{Hom}_{O(V, q)}(D, \bigwedge^3 W) = 0$.

Proof. Put $A_2 = \text{Sym}^2 U$ and $B_2 = \text{Sym}^2 U'$. The symmetric Cauchy formula gives

$$W = (A_2 \otimes B_2) \oplus \mathbb{C}\rho, \quad \rho = q^{-1},$$

where ρ is invariant. Write $T = A_2 \otimes B_2$. For any two three-dimensional vector spaces, antisymmetrization in $(A_2 \otimes B_2)^{\otimes r}$, for $r = 2, 3$, gives

$$\begin{aligned} \bigwedge^2 T &= (\text{Sym}^2 A_2 \otimes \bigwedge^2 B_2) \oplus (\bigwedge^2 A_2 \otimes \text{Sym}^2 B_2), \\ \bigwedge^3 T &= (\text{Sym}^3 A_2 \otimes \bigwedge^3 B_2) \oplus (\mathbb{S}_{(2,1)} A_2 \otimes \mathbb{S}_{(2,1)} B_2) \oplus (\bigwedge^3 A_2 \otimes \text{Sym}^3 B_2). \end{aligned}$$

For clarity, the reason for the conjugate partitions in these formulas is that tensoring a symmetric-group representation with the sign representation conjugates its partition. Schur–Weyl decomposition of the two tensor powers therefore gives exactly the displayed summands. This is the usual skew Cauchy formula, not an additional hypothesis about this particular q .

Here are all the required one-factor calculations:

functor on $\text{Sym}^2 U$	restriction to SL_2
Sym^2	$\text{Sym}^4 U \oplus \mathbb{C}$
$\bigwedge^2$	$\text{Sym}^2 U$
$\bigwedge^3$	$\mathbb{C}$
Sym^3	$\text{Sym}^6 U \oplus \text{Sym}^2 U$
$\mathbb{S}_{(2,1)}$	$\text{Sym}^4 U \oplus \text{Sym}^2 U$

(4.17)

The weights of $\text{Sym}^2 U$ are $2, 0, -2$. Taking unordered pairs or triples of these weights verifies the symmetric-power rows; taking distinct weights verifies the exterior rows. Complete reducibility identifies the resulting characters with the stated representations. Alternatively the final row follows from $A_2 \otimes \bigwedge^2 A_2 = \mathbb{S}_{(2,1)} A_2 \oplus \bigwedge^3 A_2$: the tensor product $\text{Sym}^2 U \otimes \text{Sym}^2 U$ is $\text{Sym}^4 U \oplus \text{Sym}^2 U \oplus \mathbb{C}$, and the exterior cube removes the scalar summand.

Substitution gives

$$\begin{aligned} \bigwedge^2 T &= [4, 2] \oplus [0, 2] \oplus [2, 4] \oplus [2, 0], \\ \bigwedge^3 T &= [6, 0] \oplus [2, 0] \oplus [4, 4] \oplus [4, 2] \oplus [2, 4] \oplus [2, 2] \oplus [0, 6] \oplus [0, 2]. \end{aligned}$$

Since $\bigwedge^3 W = \bigwedge^3 T \oplus (\bigwedge^2 T) \wedge \rho$, these equalities prove (4.16). The dimensions are $84 + 36 = 120 = \binom{10}{3}$. There is no summand $[0, 0]$. Finally the determinant character of $O(V, q)$, namely its representation on D , restricts to the trivial character of $SO(V, q)$. Any equivariant image of D would therefore be a nonzero $SO(V, q)$ -invariant vector, which the decomposition excludes. This tracks the determinant twist without choosing an extension of each nontrivial SO_4 summand to O_4 . $\square$

Lemma 4.10 (The contraction kernel). *For $0 \neq q \in \text{Sym}^2 V^*$, let $\kappa_q : D \otimes \text{Sym}^4 V \rightarrow \bigwedge^3 \text{Sym}^2 V$ be the map $\epsilon \otimes f \mapsto \iota_q j(\epsilon \otimes f)$. If q is singular, then*

$$\ker \kappa_q = \det V \otimes \text{Sym}^4(\text{rad } q). \quad (4.18)$$

If q is nondegenerate, its inverse is a quadratic tensor $q^{-1} \in \text{Sym}^2 V$, and

$$\ker \kappa_q = \det V \otimes \mathbb{C}(q^{-1})^2. \quad (4.19)$$

Proof. For each fixed q , the map in the statement is explicitly

$$\kappa_q : D \otimes \text{Sym}^4 V \longrightarrow \bigwedge^3 \text{Sym}^2 V, \quad \epsilon \otimes f \longmapsto \iota_q j(\epsilon \otimes f).$$

Naturality of polarization and contraction makes this map equivariant for the stabilizer of q in $GL(V)$. The source carries the determinant factor D ; the target has its ordinary exterior representation. We use that equivariance in both cases.

Suppose first that q has rank $r \in \{1, 2, 3\}$. Choose $V = A \oplus B$, where $B = \text{rad } q$ and $q|_A$ is nondegenerate. Every pure power b^4 , $b \in B$, is killed by contraction in (4.13), since $q(bx) = 0$ for all $x \in V$. Pure powers span $\text{Sym}^4 B$, so $D \otimes \text{Sym}^4 B \subset \ker \kappa_q$.

The stabilizer contains $O(A, q|_A) \times GL(B)$, the B -scaling torus, and all shears in Lemma 4.8. On the degree-one weight space

$$D \otimes A \otimes \text{Sym}^3 B$$

the product-group representation is irreducible. Indeed its untwisted factors are the standard representation of the full orthogonal group and the irreducible symmetric cube for $GL(B)$. The determinant factor restricts to the product character $\det_A \det_B$, which does not change irreducibility. For $r = 1$ the standard factor is one-dimensional; for $r = 2$ its two isotropic weight lines for SO_2 are exchanged by a reflection in O_2 , so neither is invariant under the full group. The usual standard representation is irreducible also for $r = 3$.

The restriction of κ_q to this space is not zero. Choose an orthogonal basis with $a = x_1 \in A$, $q(a, a) = 1$, and $b = x_4 \in B$, and set $\epsilon = x_1 \wedge \cdots \wedge x_4$. In the coefficient of st^3 in $\iota_q j(\epsilon \otimes (sa + tb)^4)$, only the contraction of the first factor contributes. Thus

$$4\kappa_q(\epsilon \otimes ab^3) = (bx_2) \wedge (bx_3) \wedge b^2 \neq 0.$$

The three factors are independent because multiplication by b is injective on V . Irreducibility now gives injectivity on the degree-one space. If the kernel contained any other positive A -degree, Lemma 4.8, with $m = 4$, would give a nonzero kernel vector in that degree-one space, a contradiction. This proves (4.18) for every singular rank, without a genericity condition on q .

Now let q be nondegenerate. After a change of coordinates, $q(x_i x_j) = \delta_{ij}$ and $\rho = q^{-1} = x_1^2 + x_2^2 + x_3^2 + x_4^2$. The harmonic decomposition is

$$D \otimes \text{Sym}^4 V = (D \otimes \mathcal{H}_4) \oplus (D \otimes \rho \mathcal{H}_2) \oplus (D \otimes \mathbb{C} \rho^2). \quad (4.20)$$

Here $\mathcal{H}_m$ is the kernel of the Laplacian in degree m . To check the decomposition directly, the symmetric Cauchy formula for $V = U \otimes U'$ in degree four has partitions $(4), (3, 1), (2, 2)$, giving SO_4 -types $[4, 4], [2, 2], [0, 0]$ of dimensions 25, 9, 1. Multiplication by the invariant ρ identifies the degree-two and scalar pieces; differentiation verifies the harmonic description. The three summands are irreducible and pairwise nonisomorphic for SO_4 . The determinant factor restricts trivially to this group.

The last summand is killed: as an $O(V, q)$ -representation, $D \otimes \mathbb{C} \rho^2 \simeq D$, and Lemma 4.9 proves that the target of κ_q has no such character. It remains to show that the other two restrictions are nonzero. Put

$$h_4 = x_1^4 - 6x_1^2 x_2^2 + x_2^4 \in \mathcal{H}_4, \quad h_2 = x_1^2 - x_2^2 \in \mathcal{H}_2.$$

In the basis triple $\tau = (x_1x_2) \wedge (x_1x_3) \wedge (x_1x_4)$, formula (4.14) gives

$$[\tau]\kappa_q(\epsilon \otimes h_4) = 2, \quad [\tau]\kappa_q(\epsilon \otimes \rho h_2) = \frac{2}{3}.$$

For the first equality the terms x_1^4 and $-6x_1^2x_2^2$ contribute 1, 1. For the second the terms $x_1^4, x_1^2x_3^2, x_1^2x_4^2$ contribute 1, $-1/6, -1/6$; all other terms contribute zero. These are coefficient identities, not numerical rank assumptions. Each nonzero restriction from an irreducible summand is injective, and the images of the two nonisomorphic summands cannot intersect. The kernel is therefore precisely $D \otimes \mathbb{C}\rho^2$. Equivariance transports this conclusion back to the original form q , proving (4.19). $\square$

For $z \in \wedge^4 W$, write $s(z)$ for the dimension of its smallest exterior support: the smallest subspace S with $z \in \wedge^4 S$. In any basis,

$$s(z) = 10 - \dim\{q \in W^* : \iota_q z = 0\}. \quad (4.21)$$

Indeed the vanishing of contraction by a dual basis vector is exactly the absence of all wedge monomials containing its corresponding basis vector. Applying this after a change of basis proves both the formula and the existence of the smallest support.

Proposition 4.11 (Exact support of the Schur embedding). *Let $0 \neq f \in \text{Sym}^4 V$ have essential-variable dimension d . With the determinant factor understood,*

d	condition on f	$s(j(f))$
1	arbitrary	4
2	arbitrary	7
3	arbitrary	9
4	f not the square of a nondegenerate quadratic	10
4	f the square of a nondegenerate quadratic	9

(4.22)

In particular, $s(j(f)) \geq 9$ if $d \geq 3$, and it is 10 if the projective quartic $V(f)$ is smooth.

Proof. If $d < 4$ and H is the essential-variable space, the contraction kernel lemma says that the annihilating bilinear forms are exactly those with $H \subset \text{rad } q$. They form $\text{Sym}^2(V/H)^*$, of dimension $(4-d)(5-d)/2$. No nondegenerate form can annihilate $j(f)$, since the square of a nondegenerate quadratic has four essential variables. This gives the first three entries of (4.22).

If $d = 4$, no singular form annihilates $j(f)$. A nondegenerate form does so exactly when f is proportional to its inverse square. In this case its projective class is unique: proportional squares of nonzero quadratics over $\mathbb{C}$ have proportional square roots, and inversion is projectively unique. The annihilator therefore has dimension one in the square case and zero otherwise. Formula (4.21) gives the last two entries.

A quartic with fewer than four essential variables is a cone and is singular at its vertex. A square of a quadratic is nonreduced and has vanishing gradient along that quadratic. Neither defines a smooth quartic surface, proving the last assertion. $\square$

Lemma 4.12 (The eight-dimensional secant and tangent bound). *A sum of two decomposable four-vectors in $\wedge^4 W$ has exterior support at most eight. Every vector in the affine tangent space $\wedge^3 R \wedge W$ to the Pluecker cone at $w_R \neq 0$ also has exterior support at most eight.*

Proof. For a sum, take the sum of the two supporting four-planes. For a tangent vector, choose a basis $r_1, \dots, r_4$ of R . Every element of $\wedge^3 R \wedge W$ has the form

$$\sum_{i=1}^4 r_1 \wedge \dots \wedge r_{i-1} \wedge u_i \wedge r_{i+1} \wedge \dots \wedge r_4$$

for four vectors $u_i \in W$, and is supported in $R + \langle u_1, \dots, u_4 \rangle$. $\square$

Theorem 4.13 (No exceptional recombination above three essential variables). *Let $R \in \text{Gr}(4, \text{Sym}^2 V)$ have nonzero Jacobian quartic $f_R = Cw_R$ with at least three essential variables. Then both Pw_R, Qw_R are nonzero and*

$$\dim H_{w_R} = 2, \quad \mathcal{L}_R \cap \text{Gr}(4, W) = \{[w_R]\}$$

as schemes. In particular,

$$G_4^{\text{rec}} = G_4^\circ. \quad (4.23)$$

The secant, double-point, pure-endpoint and flag-line strata of the component pencil are all disjoint from G_4° .

Proof. By (4.15) and Proposition 4.11, the vector Qw_R has support at least nine. It is nonzero and cannot be proportional to the decomposable vector w_R ; hence $Pw_R \neq 0$. On the component line, the quadratic Pluecker ideal restricts to $(b-a)H_{w_R}$, where $H_{w_R} = \langle \ell_{F,R} \rangle \subset H^0(\mathbb{P}^1, \mathcal{O}(1))$. If this space has dimension two, its ideal sheaf is $(b-a)$. If it has dimension one, the intersection is the degree-two divisor of $(b-a)\ell$, including multiplicities. If it is zero, the entire line is contained in the Grassmannian. These assertions follow directly from the ideal generated by the Pluecker quadrics; no classification of an ambient boundary is used.

If the closed pencil has a distinct second decomposable point $z = aPw_R + bQw_R$ with $a \neq b$, then $z = aw_R + (b-a)Qw_R$. Thus Qw_R is a linear combination of two decomposable vectors, contradicting Lemma 4.12. This argument includes a pure endpoint. If the residual point coincides with $[w_R]$, then the line is tangent at $[w_R]$: its restricted equations have zero first derivative there. Its direction is Qw_R modulo Cw_R , so $Qw_R \in \wedge^3 R \wedge W$, giving the same contradiction. A whole Grassmann line also has that tangent property. All alternatives to rank two are excluded.

Smoothness of Y_R implies four essential variables, so every $R \in G_4^\circ$ satisfies this conclusion. The equality in (4.23) holds as an equality of open subschemes. Here G_4° has the reduced open-subscheme structure inherited from $\text{Gr}(4, 10)$. The complement is defined by the two-by-two minors of the two-column coefficient matrix of the residual linear forms. It is a closed subscheme of finite type over $\mathbb{C}$ with no geometric points, hence is empty, regardless of possible nilpotents in that determinantal presentation. The statement for all the named exceptional strata follows at the same time. $\square$

Corollary 4.14 (An invariant location of the ambient obstruction). *On $X^\times$, the support of the rank-defect locus $\dim H_w \leq 1$ is contained in the inverse image under the Jacobian covariant of the subspace variety of binary quartics. That variety is defined set-theoretically by rank at most two of the first catalecticant $V^* \rightarrow \text{Sym}^3 V$. In particular, it produces no exceptional divisor or ramification locus in the smooth geometric family, and no pair of non-projectively-equivalent smooth webs with isomorphic full failure schemes.*

Proof. The containment is the contrapositive of Theorem 4.13; the catalecticant characterization is the essential-variable characterization above. The inverse-scheme assertion follows from the intrinsic readout and Theorem 4.1. No equality with the whole binary-Jacobian inverse image, or classification of its ambient irreducible components, is asserted. $\square$

Proof of Theorem 4.1. Theorem 4.13 proves that the rank-two condition holds on all of G_4° , rather than merely on a nonempty subopen. The coefficient-rank condition in (4.3) defines this open algebraically. Nonemptiness of the geometric locus is independently witnessed by the smooth integral web C_* of Lemma 6.1; Appendix S14 retains its exact separating quadrics and the square-web normalization.

Indeed the restricted quadrics are $(b-a)H_{w_R}$. When H_{w_R} is the full space of linear forms, those forms generate the unit ideal sheaf on the line, and the intersection has ideal sheaf $(b-a)$. It is the single reduced point $[w_R]$. Without that rank condition, the points with the same two nonzero

projective components are precisely the decomposable points $[aPw_R + bQw_R]$ with $ab \neq 0$, which proves (4.4). This is an exact characterization of the ambiguity of the *two-component invariant*; it need not describe every additional invariant of the full failure scheme on an exceptional locus. If the line is not contained in the Grassmannian, its scheme-theoretic intersection has length at most two, because the Grassmannian ideal is generated by quadrics.

An abstract isomorphism $\widehat{D}_R \simeq \widehat{D}_{R'}$ induces the graded normal-cone map of Lemma 4.7. Its commutative coefficient square and Lemma 4.4 give one $g \in \mathrm{PGL}(V)$ for which the pairs $([Pw_{R'}], [Qw_{R'}])$ and $([P(gw_R)], [Q(gw_R)])$ agree. The two components are transported by the same g ; the source-factor change acts only on the right Cauchy factors. Replace R' by $g^{-1}R'$. Its Pluecker point belongs to the pencil $\mathcal{L}_R$, and both components are nonzero. The reduced one-point intersection gives $[w_{R'}] = [w_R]$, hence $R' = R$. The algebra-orbit equivalence of Lemma 2.4 gives the assertion for B_R . $\square$

Corollary 4.15 (Separation of the Torelli packet). *On the finite etale locus of the web-to- $K3$ map in Theorem S3.3, every fibre is separated by the isomorphism class of the full failure scheme. No further restriction to a reconstruction subopen is necessary. Distinct projective web classes in one such fibre have nonisomorphic failure schemes. On the regular Reye locus, the full failure scheme therefore also determines the web's Reye congruence and its Enriques datum.*

Proof. Every web in the indicated finite etale cover belongs to $G_4^\circ = G_4^{\mathrm{rec}}$. Theorem 4.1 therefore separates all its projective web classes directly. The Reye and Enriques constructions on the regular web locus are functorial under projective equivalence, which proves the last assertion. $\square$

The result concerns the actual degree-sixteen ideal of the deepest transverse failure cone. It uses information that is lost on passing to the reduced incidence image or to the Jacobian quartic alone. It neither identifies an incidence-resolution decomposition with a primary decomposition of W_3 or W_4 , nor asserts that all higher embedded primary components have been classified. Its geometric conclusion is instead an intrinsic inverse theorem: for every smooth Jacobian web, the deeper nonreduced boundary recovers the original relation space.

5. RECOVERING THE POLARIZED RAMIFICATION VARIETY

Let $A = \mathcal{O}_Y(1)$ and $B = \mathcal{O}_{\mathbb{P}(S)}(1)$. In the graph description (3.3) write

$$Z = Y \times \mathbb{P}(S), \quad W = \mathcal{H}^* \otimes Q, \quad Q = (S \otimes \mathcal{O})/\Lambda, \quad \pi : E = \mathrm{Tot}_Z W \rightarrow Z,$$

and let $\rho : E \rightarrow Y$ be the composite projection. Constant one-dimensional vector-space factors in determinant lines below are suppressed; they change none of the isomorphism classes or section rings.

Lemma 5.1 (No extra fibrewise functions). *The natural map $\mathcal{O}_Y \rightarrow \rho_* \mathcal{O}_E$ is an isomorphism.*

Proof. The assertion is local on Y , where $\mathcal{H}$ can be trivialized. If H_0 is its fibre space, the affine bundle projection gives

$$\pi_* \mathcal{O}_E = \bigoplus_{d \geq 0} \mathrm{Sym}^d(H_0 \otimes Q^*).$$

We claim that

$$H^0(\mathbb{P}(S), \mathrm{Sym}^d(H_0 \otimes Q^*)) = 0 \quad (d > 0). \quad (5.1)$$

The injection $Q^* \hookrightarrow S^* \otimes \mathcal{O}$ embeds a global section in the constant space $\mathrm{Sym}^d(H_0 \otimes S^*)$. Such an element is a homogeneous polynomial on $\mathrm{Hom}(H_0, S)$. At every line $\Lambda \subset S$, membership in $\mathrm{Sym}^d(H_0 \otimes (S/\Lambda)^*)$ says this polynomial factors through $\mathrm{Hom}(H_0, S/\Lambda)$; it is invariant under translations by $\mathrm{Hom}(H_0, \Lambda)$. Taking the lines of a basis of S shows invariance under every

translation of $\mathrm{Hom}(H_0, S)$. In characteristic zero such a polynomial is constant, so its positive-degree part is zero. This proves (5.1). The degree-zero part is $\mathbb{C}$.

The same calculation tensors with the coordinate ring of each affine trivializing open of Y : cohomology on the projective space commutes with this flat extension from $\mathbb{C}$, and with the displayed direct sum in degree zero. Hence the pushforward is precisely $\mathcal{O}_Y$, with its natural multiplication. The local identifications glue. $\square$

Proposition 5.2 (The canonical nilradical line). *As line bundles on E ,*

$$\mathcal{N} \simeq \pi^*(A^{-1} \boxtimes B^{-1}), \quad \omega_E \simeq \pi^*(A^{-(p-1)} \boxtimes B^{-(p+e-1)}). \quad (5.2)$$

Consequently

$$\omega_E \otimes \mathcal{N}^{\otimes -(p+e-1)} \simeq \rho^* A^e. \quad (5.3)$$

Proof. The determinant defining $\hat{\Delta}$ is a section of $(\det \mathcal{K})^* \otimes \det V$. On E the exact sequence $0 \rightarrow \Lambda \rightarrow \mathcal{K} \rightarrow \mathcal{H} \rightarrow 0$ gives $\mathcal{I}_{\Delta}|_E = \det \mathcal{H} \otimes \Lambda$, up to a constant factor. Since $\det \mathcal{H} = A^{-1}$ and $\Lambda = B^{-1}$, Theorem 3.2 gives the first formula.

The determinant of W is $(\det \mathcal{H}^*)^{p-1} \otimes (\det Q)^{e-1} = A^{p-1} \boxtimes B^{e-1}$. The canonical bundle of a vector-bundle total space is $\pi^*(\omega_Z \otimes \det W^{-1})$, as follows by taking the determinant of its relative cotangent sequence. Here $\omega_Z = \mathcal{O}_Y \boxtimes B^{-p}$ because $\omega_Y = \mathcal{O}_Y$. This proves the second formula. Subtracting the two exponents gives (5.3): the B -exponent vanishes and the A -exponent is $-(p-1) + (p+e-1) = e$. $\square$

Proof of Theorem 1.2. An isomorphism of failure schemes preserves the nilradical and its annihilator, hence E , its invertible module $\mathcal{N}$, and its canonical bundle. It therefore preserves $\mathcal{L} = \omega_E \otimes \mathcal{N}^{-(p+e-1)}$ and its graded section ring. Proposition 5.2, Lemma 5.1, and the projection formula give

$$H^0(E, \mathcal{L}^n) = H^0(Y, A^{en} \otimes \rho_* \mathcal{O}_E) = H^0(Y, A^{en})$$

compatibly with multiplication. The full section ring of the ample line bundle A^e recovers (Y, A^e) by Proj. In this case it is also the e th Veronese of the homogeneous coordinate ring of a smooth hypersurface, so there is no ambiguity about the grading line bundle.

For a K3 surface the Picard group has no torsion. Here is the short argument needed: a nontrivial torsion line bundle T and its inverse have no nonzero sections, since a nonzero effective torsion divisor has zero intersection with an ample divisor, which is impossible. Riemann–Roch and Serre duality would then give $2 = \chi(T) = -h^1(T) \leq 0$, a contradiction. Thus an isomorphism carrying A^4 to $(A')^4$ carries A to A' . This recovers the quartic polarization, and its complete linear system recovers the quartic embedding up to projective equivalence.

Finally Proposition 3.1 recovers D from $\hat{D}$ by restricting to the smooth locus of the reduction. This operation is intrinsic and is preserved by every abstract scheme isomorphism. The preceding proof applies to that restriction. $\square$

Remark 5.3. For $e = 3$ the theorem recovers (Y, A^3) directly. A chosen cubic root in the Picard group need not be unique. As polarized genus-one curves, however, the possible degree-three roots are isomorphic: after choosing an origin, the map $y \mapsto t_y^* A \otimes A^{-1}$ onto $\mathrm{Pic}^0(Y)$ is multiplication by three, hence surjective. No choice of an origin is needed for the stated section-ring reconstruction. For $e = 4$ the torsion-free argument gives uniqueness without translations.

Remark 5.4 (Relation to the earlier birational argument). The compactification $\mathbb{P}_Z(\mathcal{O} \oplus W)$ is birational to $Y \times \mathbb{P}^{e(p-1)}$. Thus the earlier cancellation proof is a special case of the maximal rationally connected quotient: for a non-uniruled base, that quotient of a product with projective space is the base, birationally. See the rational-quotient construction in [14, Chapter IV, §5]. This explains the earlier abstract recovery, which is retained in the companion. The present proof

does not rely on that argument: the section ring gives the actual projective model and its line bundle, not merely a birational class.

6. NINE-DIMENSIONAL VARIATION IN THE K3 CASE

We prove Theorem S5.3. Work on the standard chart of $\text{Gr}(4, \text{Sym}^2 \mathbb{C}^4)$ where the four square coefficients are the identity. Write $x = (x_0, x_1, x_2, x_3)$ and order the six cross terms as

$$m = (x_0x_1, x_0x_2, x_0x_3, x_1x_2, x_1x_3, x_2x_3).$$

For $C \in \text{Mat}_{6 \times 4}$ set

$$q_i = x_i^2 + \sum_{k=1}^6 C_{ki} m_k, \quad J_C = (\partial q_i / \partial x_j), \quad F_C = \det J_C. \quad (6.1)$$

This parametrizes the relation space, not an additional choice of its basis: the square coefficients fix that basis on this chart.

Lemma 6.1 (An integral differential witness). *At the relation system*

$$C_* = \begin{pmatrix} 2 & -1 & 2 & 0 \\ -2 & 0 & -1 & 1 \\ 2 & -1 & -2 & 0 \\ 1 & -1 & 2 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & -2 & 1 & -2 \end{pmatrix}, \quad (6.2)$$

the differential of $C \mapsto [F_C] \in \mathbb{P}(\text{Sym}^4 \mathbb{C}^4)$ has rank 24. This relation system is basepoint-free and $V(F_{C_*})$ is smooth over $\mathbb{C}$.

Proof. Let $\text{cof}_{ij}(J)$ denote the signed cofactor. The 24 parameter derivatives are

$$G_{ik} = \left. \frac{\partial F_C}{\partial C_{ki}} \right|_{C_*} = \sum_{j=0}^3 \text{cof}_{ij}(J_{C_*}) \frac{\partial m_k}{\partial x_j}. \quad (6.3)$$

Order degree-four monomials by descending lexicographic order of their exponent tuples, and form the 35×25 integral coefficient matrix D_* with columns F_{C_*} , followed by G_{ik} with $i = 0, \dots, 3$ outermost and $k = 1, \dots, 6$ innermost. The determinant of its first 25 rows and all columns is

$$4279473148893659522379284480 \neq 0. \quad (6.4)$$

Its residue modulo 101 is 41. Thus F_{C_*} and the 24 derivatives are linearly independent. Quotienting the target tangent space by the line spanned by F_{C_*} proves the differential assertion.

For completeness, Appendix S16 specifies the full coefficient recipe and two additional finite rank certificates. The Macaulay matrix of the q_i in degree five has rank 56, the dimension of all degree-five polynomials. The analogous matrix of the four partials of F_{C_*} in degree nine has rank 220, again the entire homogeneous piece. Explicit square minors are nonzero modulo 101, with residues 44 and 79. The integer matrices therefore have full row rank over $\mathbb{Q}$. Every monomial of those degrees belongs to the respective homogeneous ideal over $\mathbb{Q}$, and hence over $\mathbb{C}$. A common projective zero is impossible. This proves basepoint-freeness and smoothness without a numerical tolerance or a probabilistic test. $\square$

Proposition 6.2. *The Jacobian map on G_4° has a 24-dimensional image in the space of embedded smooth quartics. Its image in polarized quartic K3 moduli has dimension nine. Its general fibres after projective coordinate equivalence are finite.*

Proof. Changing a basis of R multiplies the determinant by a nonzero scalar, so the projective Jacobian map is well-defined. A change of coordinates in V acts equivariantly. Lemma 6.1 and the 24-dimensional domain show that its image has dimension 24 and that the map is generically finite onto that embedded image. The locus where the differential is injective is nonempty and open; it is invariant because the map is equivariant. The smooth admissible locus is irreducible by Proposition 2.3.

The projective stabilizer of a smooth quartic K3 is finite. Indeed, contraction with its nonzero two-form identifies its tangent bundle with its cotangent bundle, and $H^0(\Omega^1) = 0$ by $H^1(\mathcal{O}) = 0$ and Hodge symmetry over $\mathbb{C}$. The stabilizer, a closed finite-type subgroup of PGL_4 , therefore has zero-dimensional Lie algebra and is finite. Its orbits have dimension 15. Two smooth quartic embeddings define the same polarized K3 precisely when they are related by PGL_4 , because their embeddings are given by the complete polarizing linear systems. The geometric quotient of the smooth quartic locus gives the quartic K3 moduli space; equivalently one may compute on a generic geometric quotient. The invariant image consequently has dimension $24 - 15 = 9$ after quotienting. Standard background on polarized K3 moduli is given in [12, Chapter 5].

To check the finiteness assertion directly, take a general embedded quartic in the 24-dimensional image where the fibre is finite. Every relation system mapping to a projectively equivalent quartic can be moved by the corresponding projective transformation into that finite fibre. Thus there are only finitely many projective equivalence classes above the general polarized point. This argument also makes clear that the assertion is generic, rather than a statement about all exceptional fibres. $\square$

Proof of Theorem S5.3. Combine Proposition 6.2 with the recovery of the polarized quartic in Theorem 1.2 and the algebra-orbit equivalence of Lemma 2.4. $\square$

Corollary 6.3. *There is a flat family of failure schemes on the corank-one open whose canonical polarized invariant has a nine-dimensional image. In particular the associated K3 surfaces are not isotrivial, and the failure schemes are not all abstractly isomorphic. Their reduced divisors become the same scheme after identifying the vector spaces S_R of rank six.*

Proof. Flatness is Corollary S1.3. A fixed abstract K3 cannot give a positive-dimensional algebraic family of polarization classes on that surface: its Picard scheme is discrete, since $H^1(\mathcal{O}) = 0$. Thus the positive-dimensional polarized image also implies non-isotriviality of the surfaces. Distinct values of the canonical polarized invariant give nonisomorphic failure schemes by Theorem 1.2. The graph description of Δ in (3.2) is independent of R . $\square$

7. THE CLASSICAL INCIDENCE SURFACE AND THE NEW INVARIANT

This section distinguishes three geometric objects associated with the same relation system. Let $e = 4$ and let $q_i(x, y)$ denote the symmetric bilinear polarization of $q_i(x)$, normalized by $q_i(x, x) = q_i(x)$. Define

$$Z_R = \{([x], [y]) \in \mathbb{P}(V^*) \times \mathbb{P}(V^*) : q_i(x, y) = 0 \text{ for all } i\}. \quad (7.1)$$

Proposition 7.1. *For $R \in G_4^\circ$, projection on the first factor identifies Z_R with Y_R . Transposition of the two factors induces a fixed-point-free involution on this K3 surface. The polarization recovered in Theorem 1.2 is the first projection's quartic polarization.*

Proof. The bilinear equations at x are $J_R(x)y = 0$, up to the common nonzero factor two. Off Y_R there is no projective solution; on Y_R the matrix has rank exactly three because Y_R is smooth. Inverting a nonzero three-minor gives a unique kernel line, with its scheme structure, and these kernel lines glue to an inverse for the projection $Z_R \rightarrow Y_R$. The equations are symmetric in x, y , so transposition acts. A fixed point would have $[x] = [y]$ and hence $q_i(x) = 0$ for every i , contrary

to basepoint-freeness. The final assertion follows from the first projection and its hyperplane bundle. $\square$

The discriminant of the web is a different embedded hypersurface. If $q_i(x) = x^t A_i x$, its equation is

$$\det\left(\sum_{i=0}^3 \lambda_i A_i\right) = 0 \quad \text{in } \mathbb{P}(R), \quad (7.2)$$

whereas the ramification equation is $\det J_R(x) = 0$ in $\mathbb{P}(V^*)$. The former varies the quadric in the web; the latter varies a point of the source of the quadratic morphism. The two determinants must not be identified simply because both have degree four. The classical construction recalled in [13, §4] relates the bilinear K3 surface to the resolution of the web's symmetroid and to a nodal Enriques quotient; see also the account of Reye congruences in [9]. Proposition 7.1 places our ramification surface in that construction by an explicit kernel-line isomorphism. No claim is made that the existence of that K3 surface, its involution, or webs of quadrics is new.

The invariant used here starts instead with the abstract multiplication-failure scheme. It obtains E as the annihilator of its nilradical, constructs $\omega_E \otimes \mathcal{N}^{-9}$ without the web or either projective space, and recovers the section ring of $\mathcal{O}_{Y_R}(4)$. The corank-two theorem further recovers an additional associated support from the square of the nilradical. These are statements about nonreduced multiplication-failure schemes, not merely about the classical incidence surface or the discriminant cycle.

8. SUBSPACE VARIETIES AND THE SUPPORT CALCULATION

The support argument has two distinct ingredients. The description of tensors by contraction flattenings, the tangent formula for a Grassmannian, and the Schur decomposition are classical. The additional calculation here is the kernel of

$$\kappa_q : \det V \otimes \text{Sym}^4 V \longrightarrow \bigwedge^3 \text{Sym}^2 V$$

for every nonzero symmetric bilinear form q , followed by its application to the two component lines recovered from the intrinsic normal cone. We give a geometric formulation that distinguishes this calculation from general rank or orbit questions.

For a vector space W define the exterior subspace variety

$$\text{Sub}_r^\wedge(W) = \{[z] \in \mathbb{P}(\bigwedge^4 W) : z \in \bigwedge^4 S \text{ for some } S \subset W, \dim S \leq r\}.$$

It is the reduced closed set defined by rank at most r of $W^* \rightarrow \bigwedge^3 W$, $q \mapsto \iota_q z$. Indeed the support-annihilator identity (4.21) identifies that rank with the smallest support dimension. Thus the minors give its support; we make no assertion here that their ideal is radical. Similarly, $\text{Sub}_d^{\text{sym}}(V) \subset \mathbb{P}(\text{Sym}^4 V)$ is the reduced closed locus of quartics in at most d essential variables. It is the first-catalecticant rank locus by [31, Proposition 1]. The spaces being measured are different: V for essential variables and $W = \text{Sym}^2 V$ for exterior support.

Proposition 8.1 (Exact reduced pullbacks of support varieties). *Let $j : \mathbb{P}(\text{Sym}^4 V) \hookrightarrow \mathbb{P}(\bigwedge^4 \text{Sym}^2 V)$ be the projectivization of j ; the one-dimensional determinant twist does not affect this map. Let $\mathcal{Q} = \{[h^2] : 0 \neq h \in \text{Sym}^2 V\}$, the closed quadratic-square locus. As reduced closed subvarieties,*

$$(j^{-1} \text{Sub}_4^\wedge(W))_{\text{red}} = \text{Sub}_1^{\text{sym}}(V), \quad (8.1)$$

$$(j^{-1} \text{Sub}_7^\wedge(W))_{\text{red}} = (j^{-1} \text{Sub}_8^\wedge(W))_{\text{red}} = \text{Sub}_2^{\text{sym}}(V), \quad (8.2)$$

$$(j^{-1} \text{Sub}_9^\wedge(W))_{\text{red}} = \text{Sub}_3^{\text{sym}}(V) \cup \mathcal{Q}. \quad (8.3)$$

In particular a quartic with at least three essential variables has image outside the second secant variety $\sigma_2(\mathrm{Gr}(4, W))$, including its limiting secants.

Proof. The support values in Proposition 4.11 give the first two equalities on complex points. They also give the third: its four-essential-variable part consists exactly of squares of nondegenerate quadratics. A degenerate quadratic and its square have the same essential-variable space, since $\partial(h^2) = 2h\partial h$ in a polynomial domain. Their squares therefore already belong to $\mathrm{Sub}_3^{\mathrm{sym}}(V)$. The locus $\mathcal{Q}$ is closed as the image of the projective morphism $[h] \mapsto [h^2]$, which has no base points. All the displayed loci are closed and have the specified reduced structures, so equality on complex points proves the assertions.

A sum of two decomposable four-vectors lies in $\mathrm{Sub}_8^\wedge(W)$. That rank locus is closed, so it contains the closure of all such secant lines. Equation (8.2) then proves the last assertion. This is an exclusion statement; it does not identify the entire intersection of the binary locus with $\sigma_2(\mathrm{Gr}(4, W))$. $\square$

8.1. The classical inputs and the additional assertion. The tangent-space identity in [30, Lemma 2.1] is precisely $\widehat{T}_{w_R} \mathrm{Gr}(4, W) = \wedge^3 R \wedge W$. Its eight-dimensional support bound is the immediate linear-algebra consequence used in Lemma 4.12. We do not claim either formula, or the general study of Grassmannian secant dimensions, as new. The issue here is the position of one specified Schur subspace relative to those secants, not the dimension of a general secant variety.

Landsberg–Weyman’s subspace varieties are defined for products of tensor factors; their Theorem 3.1 describes their equations and singularities [32]. That framework explains the use of flattening ranks as support conditions. It is not a theorem about the contraction of our determinant-twisted quartic submodule, nor does it identify the reduced pullbacks (8.1)–(8.3). The exterior rank identity needed here was proved directly, so no scheme-theoretic conclusion for alternating tensors is imported from the tensor-product statement.

The occurrence of $\mathbb{S}_{(5,1,1,1)}V$ is a plethysm calculation, not a novelty claim. Its multiplicity is one in $\wedge^4 \mathrm{Sym}^2 V$, whose two highest weights are $(4, 3, 1, 0)$ and $(5, 1, 1, 1)$. For example its character identity is

$$[t^4] \prod_{i \leq j} (1 + tx_i x_j) = s_{(4,3,1,0)}(x) + s_{(5,1,1,1)}(x).$$

The highest-weight and exterior-duality verification is given in Lemma 4.4; polarization fixes the normalization of its unique quartic embedding. The skew Cauchy and SL_2 character calculations used for the scalar kernel are likewise standard representation operations, expanded in Lemma 4.9 rather than cited as a new decomposition method.

The precise additional assertions proved here are the all-ranks contraction kernels of Lemma 4.10, the resulting exact support table, and the reconstruction consequence of combining that table with the intrinsic common-coordinate readout. Exterior support is neither the least number of decomposable summands nor a classification of $GL(W)$ -orbits in $\wedge^4 W$. The reduced support pullbacks above, and the containment of the ambient reconstruction obstruction over binary quartics, must not be mistaken for either such classification. The primary comparisons just made establish which cited tools are classical; they are not an exhaustive historical priority certificate. In particular the six-way theorem comparison with the unread Ballico source [2] remains separately unresolved.

REFERENCES

- [1] E. Arrondo and I. Sols, *On congruences of lines in the projective space* (Chapter 6 written in collaboration with M. Pedreira), Mém. Soc. Math. France (N.S.) **50** (1992), 1–96. doi:10.24033/msmf.363.

- [2] E. Ballico, On the failure locus of higher order properties of embeddings in projective spaces, *Math. Nachr.* **163** (1993), 5–13. doi:10.1002/mana.19931630102.
- [3] E. Ballico, On the failure cycles for the quadratic normality of a projective variety, *Pacific J. Math.* **172** (1996), 307–313.
- [4] D. Bayer and D. Eisenbud, Ribbons and their canonical embeddings, *Trans. Amer. Math. Soc.* **347** (1995), 719–756. doi:10.1090/S0002-9947-1995-1273472-3.
- [5] W. Bruns and A. Conca, Gröbner bases and determinantal ideals, in *Commutative Algebra, Singularities and Computer Algebra*, NATO Sci. Ser. II Math. Phys. Chem. 115, Kluwer, 2003, 9–66; author preprint arXiv:math/0302058.
- [6] W. Bruns and U. Vetter, *Determinantal Rings*, Lecture Notes in Mathematics 1327, Springer, 1988.
- [7] F. Cossec, Reye congruences, *Trans. Amer. Math. Soc.* **280** (1983), 737–751. doi:10.1090/S0002-9947-1983-0716848-4.
- [8] C. De Concini, D. Eisenbud, and C. Procesi, Young diagrams and determinantal varieties, *Invent. Math.* **56** (1980), 129–165. doi:10.1007/BF01392548.
- [9] I. Dolgachev and S. Kondō, *Enriques Surfaces II*, Springer, 2025, chapter *Reye Congruences*, 67–141. doi:10.1007/978-981-96-1513-1_2.
- [10] I. Dolgachev and I. Reider, On rank 2 vector bundles with $c_1^2 = 10$ and $c_2 = 3$ on Enriques surfaces, in *Algebraic Geometry*, Lecture Notes in Mathematics 1479, Springer, 1991, 39–49. doi:10.1007/BFb0086262.
- [11] J.-M. Drézet, Primitive multiple schemes, *Eur. J. Math.* **7** (2021), 985–1045. doi:10.1007/s40879-020-00447-4.
- [12] D. Huybrechts, *Lectures on K3 Surfaces*, Cambridge Studies in Advanced Mathematics 158, Cambridge University Press, 2016. doi:10.1017/CBO9781316594193.
- [13] C. Ingalls and A. Kuznetsov, On nodal Enriques surfaces and quartic double solids, *Math. Ann.* **361** (2015), 107–133. doi:10.1007/s00208-014-1066-y.
- [14] J. Kollár, *Rational Curves on Algebraic Varieties*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 32, Springer, 1996, Chapter IV, §5.
- [15] N. Manolache, Cohen–Macaulay nilpotent schemes, arXiv:math/0312514 (2003).
- [16] G. Martin, G. Mezzedimi, and D. C. Veniani, Nodal Enriques surfaces are Reye congruences, *J. Reine Angew. Math.* **808** (2024), 49–65. doi:10.1515/crelle-2023-0092.
- [17] W. Fulton, *Intersection Theory*, 2nd ed., Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 2, Springer, 1998, §5.1.
- [18] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics 129, Springer, 1991.
- [19] The Stacks Project Authors, *The Stacks Project*, Section 31.20, “The normal cone of an immersion,” Tag 062Z, <https://stacks.math.columbia.edu/tag/062Z>.
- [20] The Stacks Project Authors, *The Stacks Project*, Section 38.21, “Flattening stratifications,” Tag 052F, <https://stacks.math.columbia.edu/tag/052F>.
- [21] The Stacks Project Authors, *The Stacks Project*, Section 31.7, “Relative assassin,” Tag 05AS, <https://stacks.math.columbia.edu/tag/05AS>.
- [22] The Stacks Project Authors, *The Stacks Project*, Section 37.18, “Flat modules and relative assassins,” Tag 0GSF; in particular Lemma 37.18.5, Tag 0GSJ.
- [23] The Stacks Project Authors, *The Stacks Project*, Section 38.18, “A criterion for purity,” Tag 05L2.
- [24] The Stacks Project Authors, *The Stacks Project*, Section 37.25, “Relative assassins,” in particular Lemma 37.25.5, Tag 05KR, <https://stacks.math.columbia.edu/tag/05KR>.
- [25] W. Bruns and W. V. Vasconcelos, Minors of symmetric and exterior powers, *J. Pure Appl. Algebra* **179** (2003), 235–240. doi:10.1016/S0022-4049(02)00115-9; arXiv:math/0111174.
- [26] S. Caracciolo, A. D. Sokal, and A. Sportiello, Algebraic/combinatorial proofs of Cayley-type identities for derivatives of determinants and pfaffians, *Adv. Appl. Math.* **50** (2013), 474–594. doi:10.1016/j.aam.2012.12.001; arXiv:1105.6270.
- [27] The Stacks Project Authors, *The Stacks Project*, Lemma 10.118.2, “Generic freeness,” Tag 051S, <https://stacks.math.columbia.edu/tag/051S>.
- [28] The Stacks Project Authors, *The Stacks Project*, Sections 37.26–37.27, “Reduced fibres” and “Irreducible components of fibres,” <https://stacks.math.columbia.edu/tag/0553>.
- [29] The Stacks Project Authors, *The Stacks Project*, Proposition 35.5.2 (Tag 023T) and Section 35.6 (Tag 0CDQ), fpqc and Galois descent of quasi-coherent modules, accessed September 23, 2026. <https://stacks.math.columbia.edu/tag/023T>.
- [30] A. Boralevi, A note on secants of Grassmannians, *Rend. Istit. Mat. Univ. Trieste* **45** (2013), 67–72. <https://rendiconti.dmi.units.it/volumi/45/011.pdf>.

- [31] E. Carlini, Reducing the number of variables of a polynomial, in *Algebraic Geometry and Geometric Modeling*, Springer, 2006, 237–247; author preprint arXiv:math/0507531, Proposition 1.
- [32] J. M. Landsberg and J. Weyman, On the ideals and singularities of secant varieties of Segre varieties, author manuscript (January 2007), Definition 1 and Theorem 3.1. <https://people.tamu.edu/~jml/1-07LWsecseg.pdf>.
- [33] The Stacks Project Authors, *The Stacks Project*, Section 29.36, Unramified morphisms, Tag 02G3, accessed September 23, 2026. <https://stacks.math.columbia.edu/tag/02G3>.